Two players take turns. Whoever takes the last counter wins, and a player who cannot move while counters are left loses. On a track, the token starts on the number of counters and moves toward 0 by the number of counters taken, and whoever puts it on 0 wins. When you play, two of you are the players and the third is the referee, and you switch roles after every game.

Problem 7: For each rule, colour every square where you would like to leave the token for your opponent.

Take 1 or 4 counters.

0	1	2	3	4	5	6	7	8	9	10
	20	19	18	17	16	15	14	13	12	11
	21	22	23	24	25	26	27	28	29	30

Take 2 or 3 counters.

0	1	2	3	4	5	6	7	8	9	10
	20	19	18	17	16	15	14	13	12	11
	21	22	23	24	25	26	27	28	29	30

Take 1, 3 or 5 counters.

0	1	2	3	4	5	6	7	8	9	10
	20	19	18	17	16	15	14	13	12	11
	21	22	23	24	25	26	27	28	29	30

Problem 8: With moves of 1, 3 or 4, does the pattern of coloured squares keep going the same way forever? Explain.

Problem 9: Choose which moves are allowed so that the squares you would like to leave are exactly 0, 5, 10, 15, and so on. Then find as many different choices as you can that make them exactly 0, 3, 6, 9, and so on. Explain how you can tell which choices work.

Problem 10: Now whoever takes the last counter loses. Take 1 or 2 counters on each turn. Play each pile until you are sure whether you would rather go first or second, and circle your choice. If you would rather go first, write how many counters you take on your first turn.

Pile of 1	first	second	take _____
Pile of 3	first	second	take _____
Pile of 5	first	second	take _____
Pile of 7	first	second	take _____
Pile of 8	first	second	take _____

Problem 11: Whoever takes the last counter still loses, so whoever puts the token on 0 loses. For each rule, colour every square where you would like to leave the token for your opponent.

Take 1 or 2 counters.

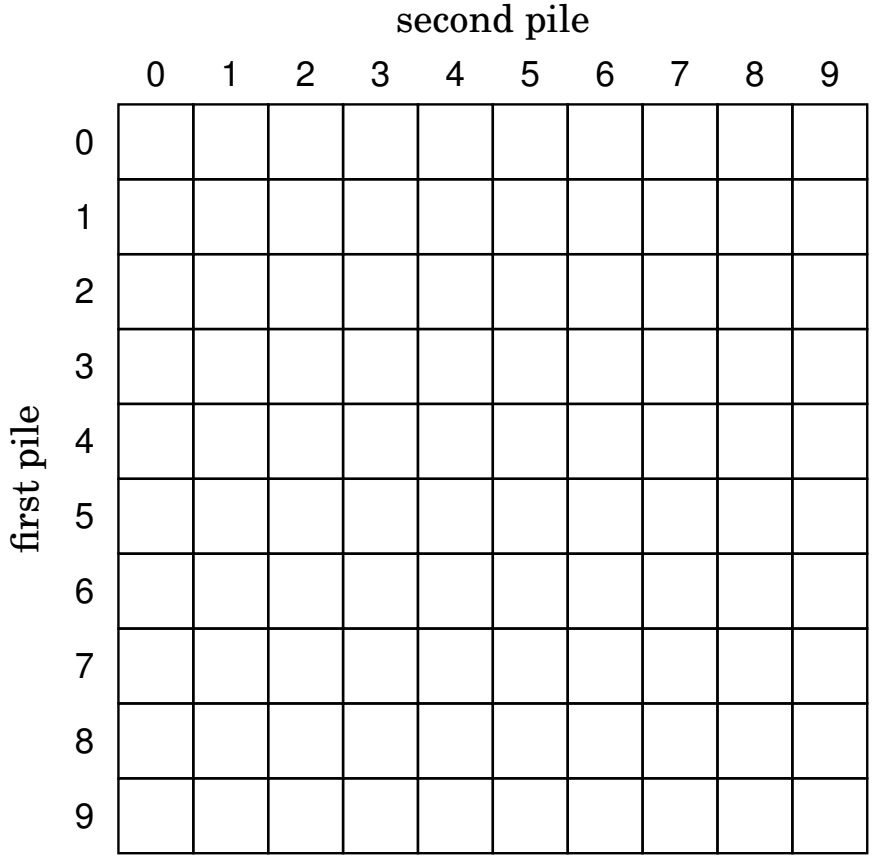
0	1	2	3	4	5	6	7	8	9	10
	20	19	18	17	16	15	14	13	12	11

Take 1, 2 or 3 counters.

0	1	2	3	4	5	6	7	8	9	10
	20	19	18	17	16	15	14	13	12	11

Problem 12: Now there are two piles. On your turn, take 1 or 2 counters from one pile, whichever you choose. Whoever takes the last counter on the table wins. Play piles of 7 and 7. Would you rather go first or second? Explain why your way of playing wins whatever your opponent does.

Problem 13: In the two-pile game of Problem 12, colour every square of the chart where you would like to leave the two piles for your opponent. Then decide whether you would rather go first or second with piles of 20 and 11, and explain.



Problem 14: Kim says that for any list of allowed moves, such as 2, 5 or 6, the coloured squares on a long enough track end up repeating. Is Kim right? Explain.

Take 2, 5 or 6 counters.

0	1	2	3	4	5	6	7	8	9	10
	20	19	18	17	16	15	14	13	12	11
	21	22	23	24	25	26	27	28	29	30